Masterschool

Personalizing Rewards to Drive Retention at TravelTide

Mastery Project — Customer Segmentation & Perk Assignment Strategy

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Problem & Goal

TravelTide faces a retention challenge: users book once and do not return. The objective is to launch a personalized rewards program that assigns each customer the perk they are most likely to value.

Cohort Definition

Users with **8 or more sessions** from January 5, 2023 onward, ensuring sufficient behavioral history for meaningful analysis.

Available Perks

- Free Checked Bag
- No Cancellation Fee
- Exclusive Discounts
- 1 Night Free Hotel
- Free Hotel Meal

Four tables, one clean analytical base

Data Sources

The foundation of the analysis is built on four core tables joined via SQL at the session level:

```
users JOIN sessions
LEFT JOIN flights
LEFT JOIN hotels
```

Cohort Filter

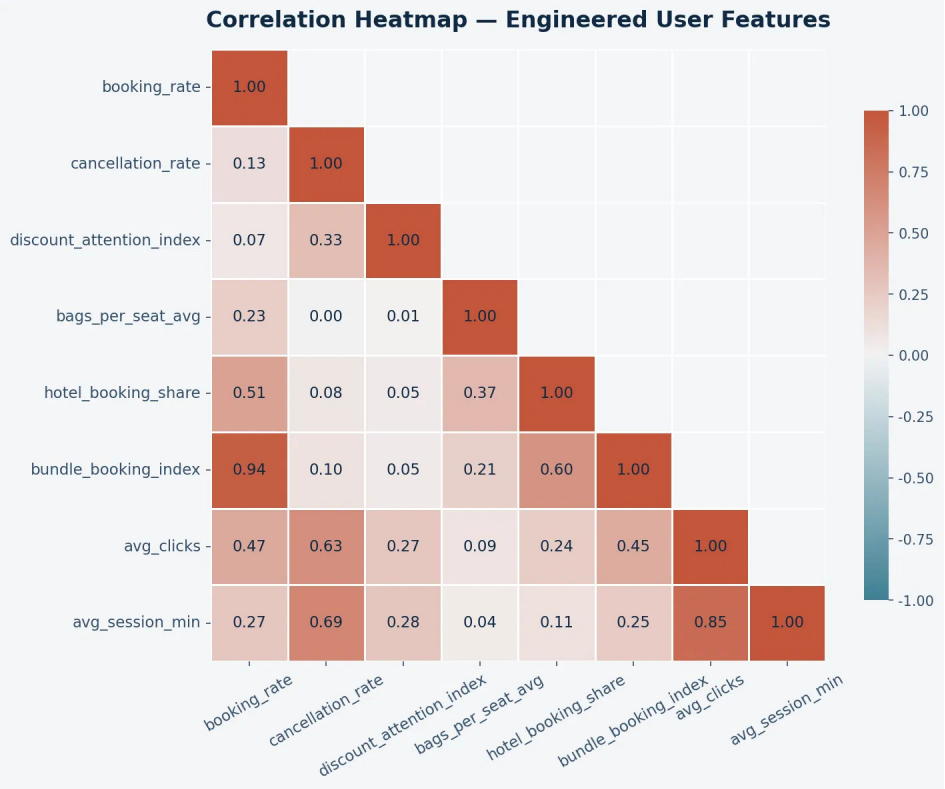
- **Minimum 8 sessions** per user to ensure sufficient behavioral history.
- **Recent activity:** Sessions from January 5, 2023 onward.

Cleaning Decisions

- **Type Conversion:** Boolean columns (married, has_children, flight_booked, hotel_booked, cancellation) converted to integers for modeling.
- **Session Duration:** Computed from session_start and session_end in minutes, clipped at 0 to handle anomalies.
- **Data Integrity:** Invalid hotel nights corrected using check-in and check-out timestamps.
- **Outlier Handling:** IQR winsorization (k=3) applied to page_clicks, session_minutes, base_fare_usd, hotel_per_room_usd, and nights to prevent clustering distortion.

Eight behavioral metrics capture what drives perk preference

FEATURE	FORMULA	BUSINESS MEANING
booking_rate	`total_bookings / total_sessions`	How often sessions convert into bookings
cancellation_rate	`cancellations / total_bookings`	Affinity for flexibility / cancellation protection
discount_booking_rate	`discounted bookings / total_bookings`	Tendency to book when a discount is present
discount_attention_index	`(flight_discounts_seen + hotel_discounts_seen) / total_sessions`	Discount exposure and responsiveness
travel_intensity_index	`total_trips / customer_age_months`	How frequently the user travels relative to tenure
bags_per_seat_avg	`avg checked_bags / avg seats`	Luggage behavior — proxy for Free Checked Bag affinity
hotel_booking_share	`hotels_booked / total_bookings`	Hotel-heavy vs flight-heavy users
bundle_booking_index	`sessions with both flight + hotel booked / total_sessions`	Tendency to book complete travel packages



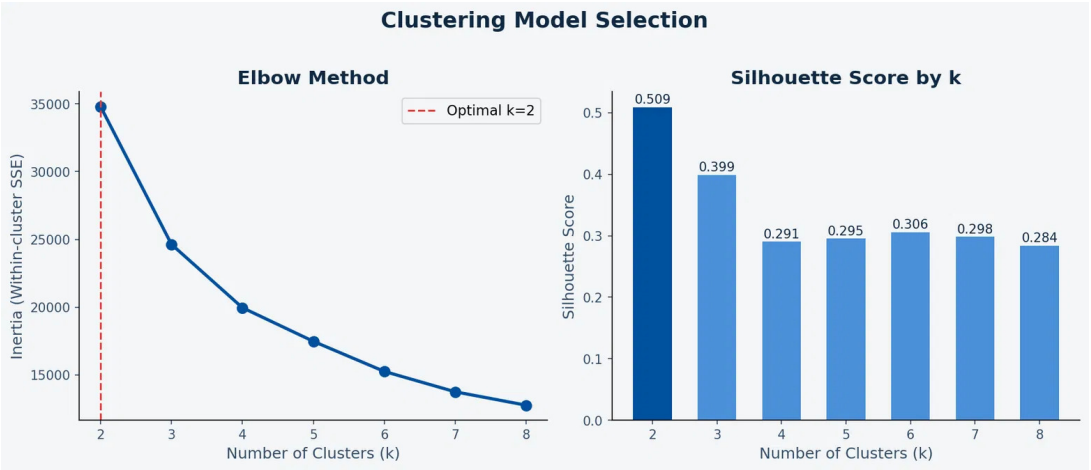
Exploratory clustering reveals two broad behavioral groups

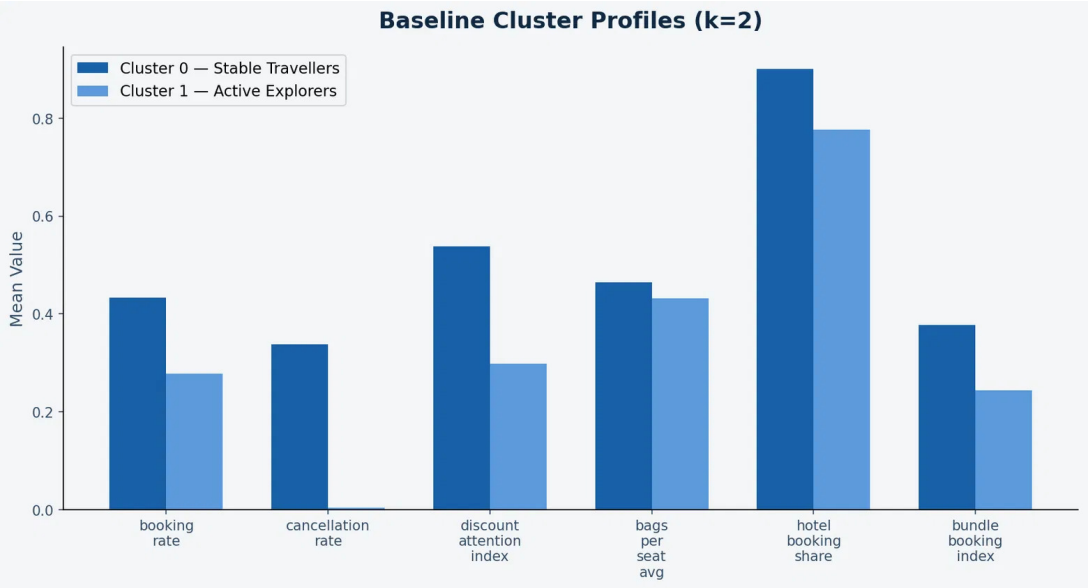
Statistical Optimum: Both the Elbow Method and Silhouette Score indicate that **k=2** provides the best mathematical separation of users.

Cluster 0 (Stable Travellers): Higher booking rates, strong preference for hotels, and tendency to book flight+hotel bundles.

Cluster 1 (Active Explorers): Higher cancellation rates, highly responsive to discounts, and more page clicks per session.

⚠ Business Limitation: While k=2 is statistically optimal, it is insufficient to effectively assign 5 distinct perks. A more granular scoring approach is required.





Every customer receives one concrete, data-driven perk

Affinity Scoring Logic

Free Checked Bag

`0.6 × bags_per_seat + 0.4 × booking_rate`

No Cancellation Fee

`0.7 × cancellation_rate + 0.3 × booking_rate`

Exclusive Discounts

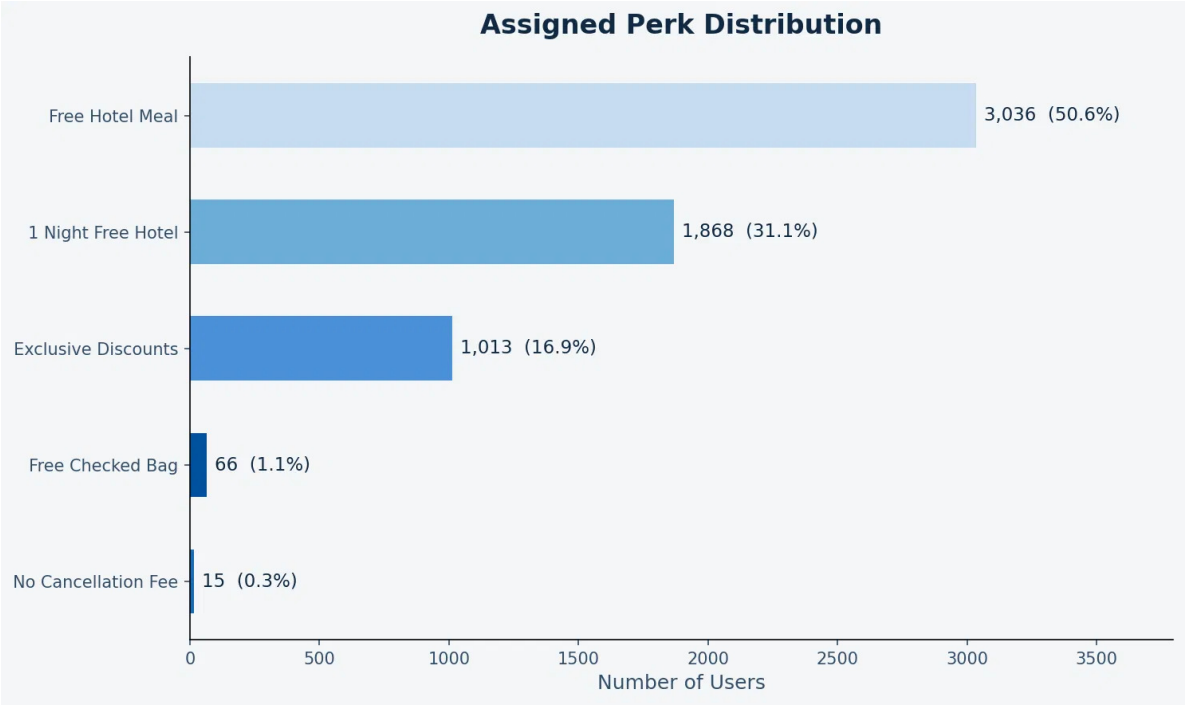
`0.7 × discount_attention + 0.3 × booking_rate`

1 Night Free Hotel

`0.6 × bundle_booking + 0.4 × hotel_share`

Free Hotel Meal

`0.6 × hotel_share + 0.4 × avg_session_min`



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Thank You

Ready to launch the personalized rewards program and drive retention.

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Next Steps & Recommendations

- **Launch A/B Test:** Roll out assigned perks to 80% of the cohort, keeping 20% as control to measure lift.
- **Monitor Margin Impact:** Track perk redemption costs vs revenue from returning users.
- **Refine Scoring Weights:** Update affinity scoring quarterly using redemption data.

KPIs to Measure Success

- **Retention Rate:** % of users who make a second booking within 90 days.
- **Perk Redemption Rate:** Frequency of perk usage at checkout.
- **Incremental Revenue per User:** Difference in LTV between treatment and control.